EECE 5554 — Lab 4

Dead Reckoning Navigation with IMU and Magnetometer

Shyam Sreenivasan

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Contents

1 Chapter 1: Yaw (Heading) Estimation

1.1 Magnetometer Calibration

Raw magnetometer readings are distorted by two effects inherent to the vehicle environment:

- **Hard-iron distortion** — permanent magnetic sources near the sensor (car body, electronics, motor) add a constant offset to every reading. The X-Y scatter plot of the magnetometer traces a circle that is displaced from the origin.
- **Soft-iron distortion** — magnetically permeable materials (steel frame) distort the local field, stretching and rotating the circle into an ellipse.

Together these produce a shifted, rotated ellipse rather than a unit circle centered at the origin. Calibration was performed using the `circle_bag` dataset, collected by driving 4–5 complete circles so that the full ellipse was traced.

Calibration Procedure

1. **Ellipse fitting.** The general conic $ax^2 + bxy + cy^2 + dx + ey = 1$ was fitted to all (m_x, m_y) pairs using ordinary least squares on the 5×1 coefficient vector.
2. **Hard-iron offset.** The ellipse center was derived analytically by setting the partial derivatives of the conic to zero:

$$\begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix} \begin{bmatrix} c_x \\ c_y \end{bmatrix} = \begin{bmatrix} -d \\ -e \end{bmatrix}$$

   Solving gives the hard-iron offset $(c_x, c_y) = (+0.048, +0.107)$ Gauss.

3. **Soft-iron correction.** In centered coordinates the ellipse is $\mathbf{x}^T Q \mathbf{x} = 1$. The correction matrix $W = V\sqrt{\Lambda}V^T$ (eigendecomposition of Q) maps any point on the ellipse to the unit circle:

$$(W\mathbf{x})^T (W\mathbf{x}) = \mathbf{x}^T W^T W \mathbf{x} = \mathbf{x}^T Q \mathbf{x} = 1$$

   The resulting matrix was nearly diagonal:

$$W = \begin{bmatrix} 4.549 & 0.047 \\ 0.047 & 4.938 \end{bmatrix}$$

   The small off-diagonal confirms the ellipse axes are nearly aligned with the sensor body axes.

After calibration the radius variation dropped to 5.97% and the axis ratio to 1.16 (1.0 = perfect circle), indicating a good but not perfect correction — expected given real-world magnetic interference throughout the drive.

Calibration Plots

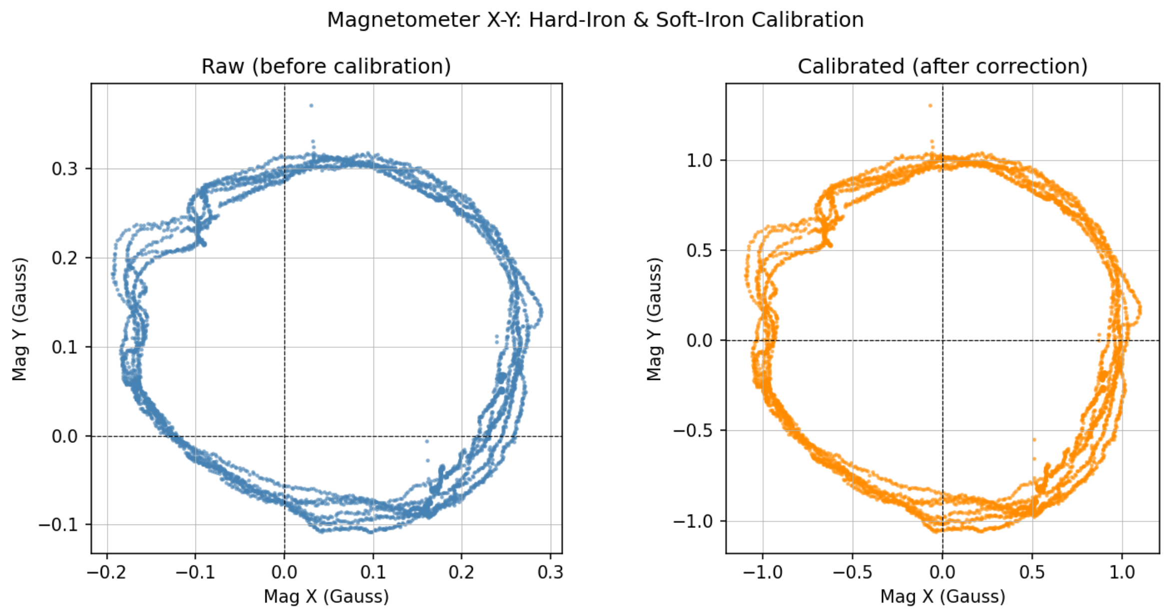

Figure 1: Magnetometer X-Y scatter before (left) and after (right) hard-iron and soft-iron calibration. The displaced ellipse is corrected to a circle centered at the origin.

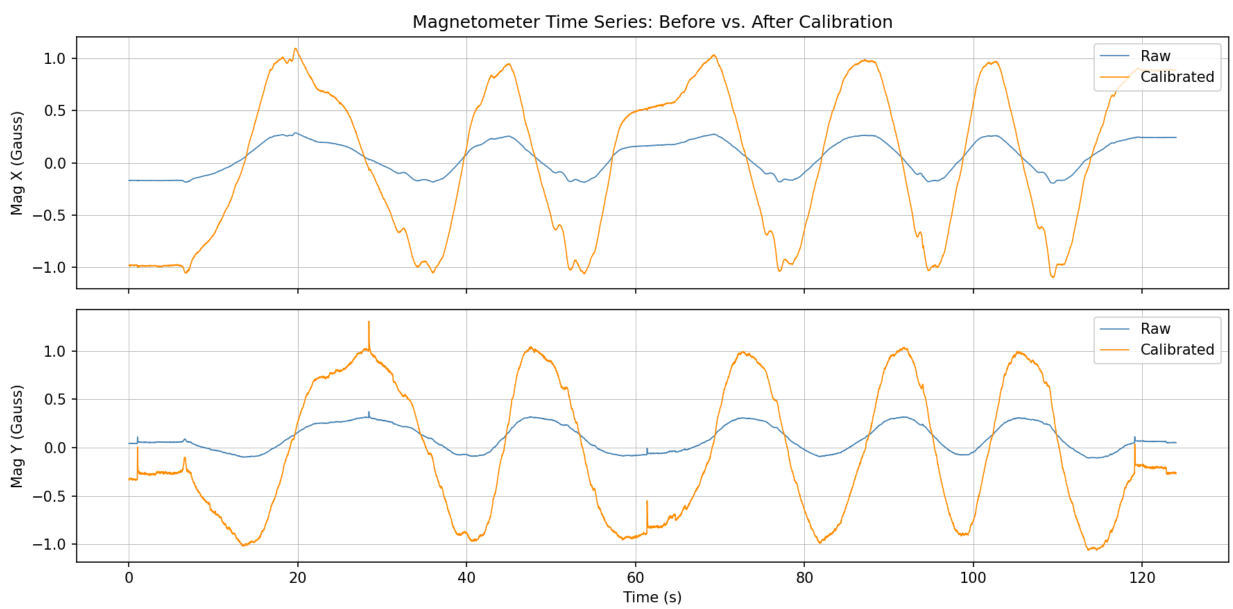

Figure 2: Magnetometer X and Y time series before and after calibration. The calibrated signals are zero-mean and have larger amplitude, each full cycle corresponding to one lap around the circle.

1.2 Yaw Estimation from the Driving Bag

Four independent heading estimates were computed from the `adventure_bag` dataset and compared.

Method 1 — Magnetometer Yaw

The calibrated magnetometer readings are converted to a heading angle using:

$$\psi_{\text{mag}}[k] = \text{atan2}(m_{y,\text{cal}}[k],\ m_{x,\text{cal}}[k]) \quad [\text{degrees}]$$

This gives an absolute heading relative to magnetic north. Because the calibrated data lies on a circle centered at the origin, the angle is geometrically correct. The signal is noisy — sensitive to vibration and nearby magnetic disturbances.

Method 2 — Gyro-Integrated Yaw

The yaw rate $\dot{\psi}$ from `gyro_z` (rad/s) was integrated using the trapezoidal rule:

$$\psi_{\text{gyro}}[k] = \psi_{\text{gyro}}[k-1] + \tfrac{1}{2}\big(\dot{\psi}[k] + \dot{\psi}[k-1]\big)\,\Delta t$$

A constant gyro bias was estimated from the first 10 seconds of stationary data and subtracted before integration. The estimated bias was 0.000461 rad/s. The signal is smooth but accumulates drift over time.

Method 3 — Complementary Filter

Both yaw signals are unwrapped to remove ±180° discontinuities. A 4th-order Butterworth filter is then applied separately, and the outputs are summed:

$$\psi_{\text{CF}}(t) = \text{LPF}\{\psi_{\text{mag}}(t)\} + \text{HPF}\{\psi_{\text{gyro}}(t)\}$$

A cutoff frequency of $f_c = 0.1$ Hz was selected by inspecting the raw yaw time series: magnetometer noise and disturbance spikes occur above ∼0.5 Hz while meaningful heading variation during urban driving is well below 0.1 Hz. Zero-phase `filtfilt` implementation eliminates phase distortion.

Method 4 — VectorNav Euler Yaw

The onboard VectorNav EKF yaw (extracted from the raw VNYMR string) is used here only as a reference for comparison. It is not used in any intermediate calculation.

Yaw Plots

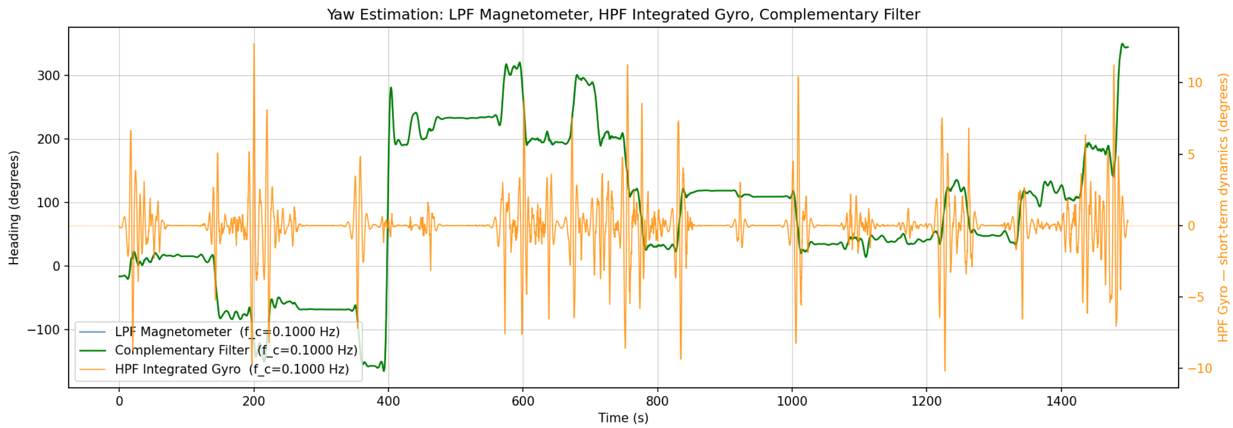

Figure 3: Low-pass filtered magnetometer (blue), high-pass filtered integrated gyro (orange, right axis), and complementary filter output (green) over the full drive. The HPF gyro captures short-term dynamics while the LPF mag provides the long-term reference.

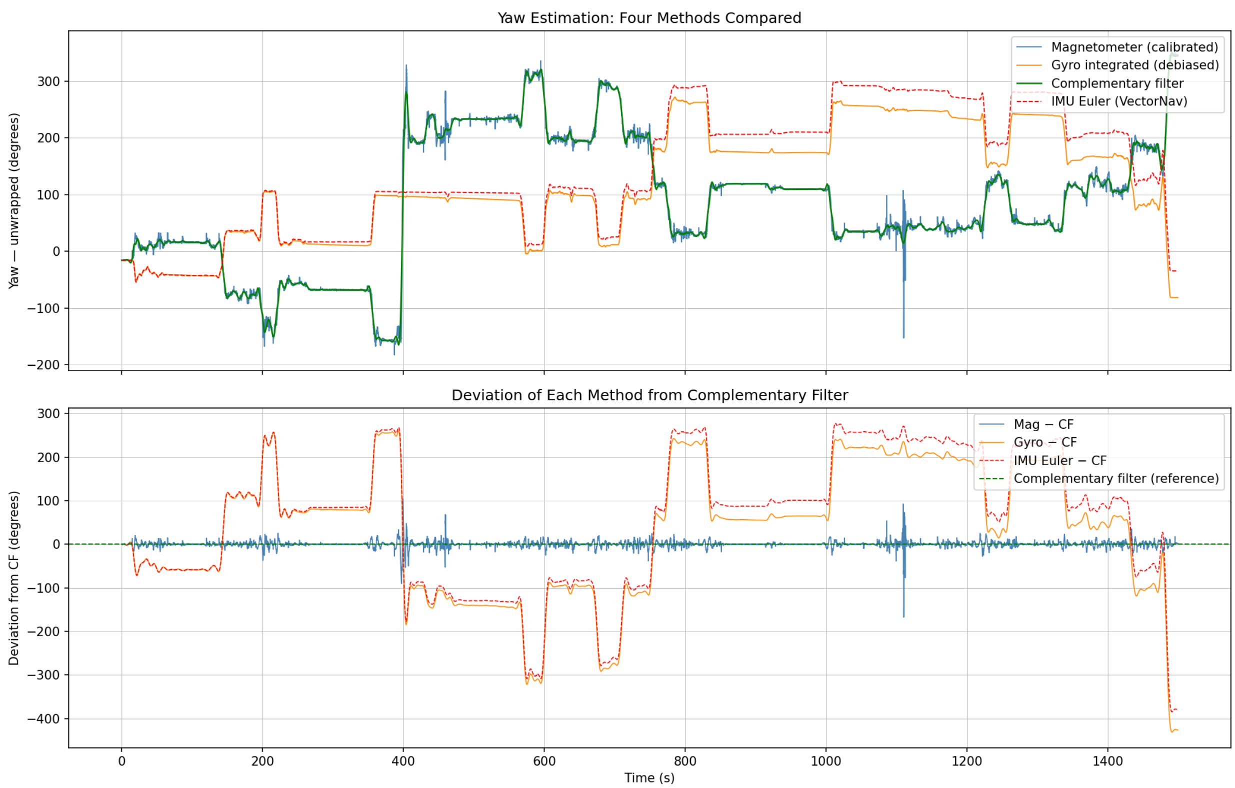

Figure 4: Comparison of all four yaw estimation methods (top) and deviation of each from the complementary filter (bottom). The gyro and VectorNav estimates track each other closely; the magnetometer shows significant noise and disturbance spikes.

1.3 Questions — Chapter 1

Q1. How did you calibrate the magnetometer? What were the sources of distortion, and how do you know?

Calibration was performed in two steps using data from the circle drive. First, a general conic ellipse was fitted to the raw (m_x, m_y) scatter via least squares. The ellipse center, derived analytically from the conic coefficients, gives the hard-iron offset $(+0.048, +0.107)$ Gauss — indicating a constant magnetic bias from the vehicle body. Second, a 2×2 soft-iron correction matrix W was constructed via eigendecomposition of the ellipse shape matrix; applying W to the centered data maps the ellipse to a unit circle. The sources of distortion are confirmed by the shape of the raw data: the displaced center is evidence of hard-iron bias, and the elongated (non-circular) shape is evidence of soft-iron distortion.

Q2. How did you convert raw magnetometer readings to angles?

After applying the calibration, the heading angle is computed as:

$$\psi = \text{atan2}(m_{y,\text{cal}},\ m_{x,\text{cal}})$$

This works because calibration places the data on a circle centered at the origin. For any point on such a circle the two-argument arctangent returns the correct angle relative to the positive X-axis, which corresponds to the sensor's forward axis and therefore the vehicle heading relative to magnetic north.

Q3. How did you use the complementary filter? What components, what cutoff frequency? Show the filter equation.

Both yaw signals were first unwrapped to remove $\pm 180°$ discontinuities. A 4th-order Butterworth filter was then applied separately to each:

$$\psi_{\text{CF}}(t) = \underbrace{\text{LPF}\{\psi_{\text{mag}}(t)\}}_{\text{slow, drift-free}} + \underbrace{\text{HPF}\{\psi_{\text{gyro}}(t)\}}_{\text{fast, low-noise}}$$

A cutoff frequency of $f_c = 0.1$ Hz was used for both filters. This value was selected by inspecting the raw yaw signals: the magnetometer contains rapid noise and disturbance spikes above $\sim$0.5 Hz, while the meaningful heading variation during urban driving occurs below 0.1 Hz. Gyro drift accumulates at DC and very low frequencies, which the high-pass filter removes. The filtfilt (zero-phase) implementation eliminates phase distortion.

Q4. Which yaw estimate(s) would you trust for navigation, and why?

The complementary filter is the most trustworthy for navigation. The magnetometer alone is too noisy and is severely corrupted by magnetic disturbances (e.g. at $t \approx 750$ s, likely a rail line). The gyro integral is smooth and consistent with the VectorNav reference but accumulates drift over long durations. The complementary filter combines both: it inherits the gyro's smoothness for short-term dynamics and the magnetometer's absolute reference for long-term drift correction. The close agreement between the CF and the VectorNav EKF yaw (which uses its own internal sensor fusion) validates this approach. For a 25-minute drive the gyro-only estimate is also reasonable given the small bias (0.000461 rad/s), but would become unreliable over longer durations without the magnetometer correction.

2 Chapter 2: Forward Velocity Estimation

2.1 Method 1 — Accelerometer Integration

Forward velocity was estimated by integrating `accel_x` (the IMU's forward body-frame axis) using the trapezoidal rule:

$$v[k] = v[k-1] + \tfrac{1}{2}\big(a_x[k] + a_x[k-1]\big)\,\Delta t$$

Adjustment — gravity removal. The IMU was mounted on the dashboard at a pitch angle of 3.207°, causing gravity to project onto the forward axis:

$$a_{x,\text{gravity}} = g\sin(\theta_{\text{pitch}}) = 9.81 \times \sin(3.207°) = 0.549\ \text{m/s}^2$$

This time-varying component was removed using the VectorNav pitch angle at each timestep:

$$a_{x,\text{adj}}[k] = a_x[k] - g\sin\!\big(\theta_{\text{pitch}}[k]\big)$$

After gravity removal, the residual constant bias was 0.00017 m/s² — negligible. The adjusted velocity was then integrated from the corrected acceleration.

Adjustment — linear drift correction. Even after gravity and bias removal, a small residual error integrates into a slow linear drift. Since the vehicle began and ended at rest at the same location, the true velocity at $t = T$ should be 0 m/s. A linear ramp equal to the final integrated velocity was subtracted:

$$v_{\text{final}}[k] = v_{\text{adj}}[k] - v_{\text{adj}}[T] \cdot \frac{k}{N-1}$$

After this correction the velocity returns to near zero at the end of the recording and stays within 0–23 m/s during driving, consistent with GPS speed.

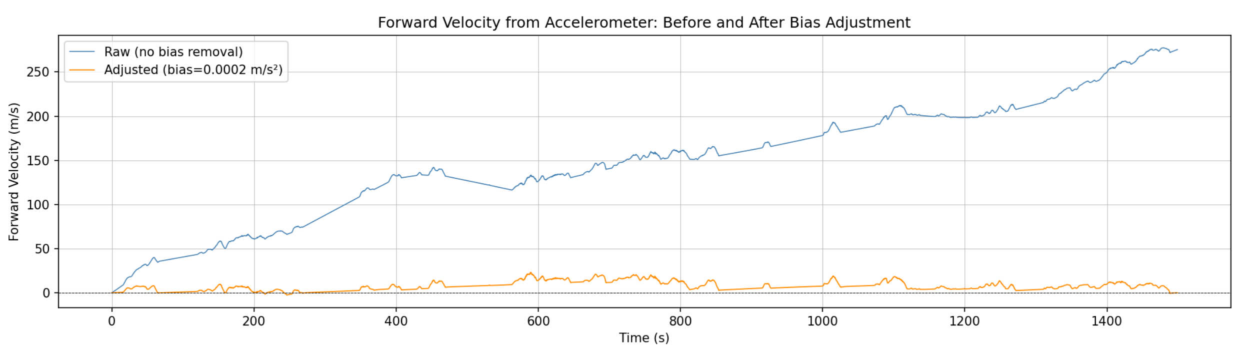

Figure 5: Forward velocity from accelerometer before (raw, blue) and after (adjusted, orange) gravity and bias correction. The raw signal drifts to +250 m/s due to the 0.549 m/s^2 gravity leak. After correction the velocity stays near zero but still drifts slowly due to unavoidable integration error.

2.2 Method 2 — GPS Velocity

Speed was estimated by finite-differencing consecutive UTM positions:

$$v_{\mathrm{GPS}}[k] = \frac{\sqrt{(\Delta E)^2 + (\Delta N)^2}}{\Delta t}$$

This gives speed magnitude at approximately 1 Hz. GPS velocity correctly captures city driving behavior: speeds of 0–10 m/s with clear stops at traffic lights.

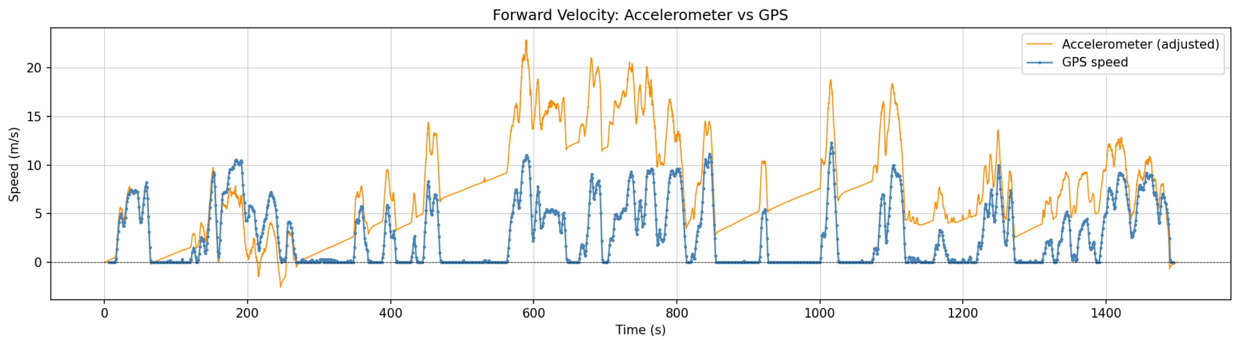

Figure 6: Adjusted accelerometer velocity (blue) vs. GPS speed (orange). GPS captures realistic city driving speeds (0–10 m/s) with stops clearly visible. The accelerometer velocity matches the general magnitude early in the drive but drifts negatively over time due to integration error.

2.3 Cross-check: $\omega \cdot \dot{X}$ vs $\ddot{y}_{\mathbf{obs}}$

Under the model assumptions (no lateral skid, IMU at center of mass), the lateral acceleration observed by the IMU should equal the centripetal term:

$$\ddot{y}_{\mathrm{obs}} = \omega \cdot \dot{X}$$

where $\omega =$ `gyro_z` and $\dot{X} =$ `v_adj`.

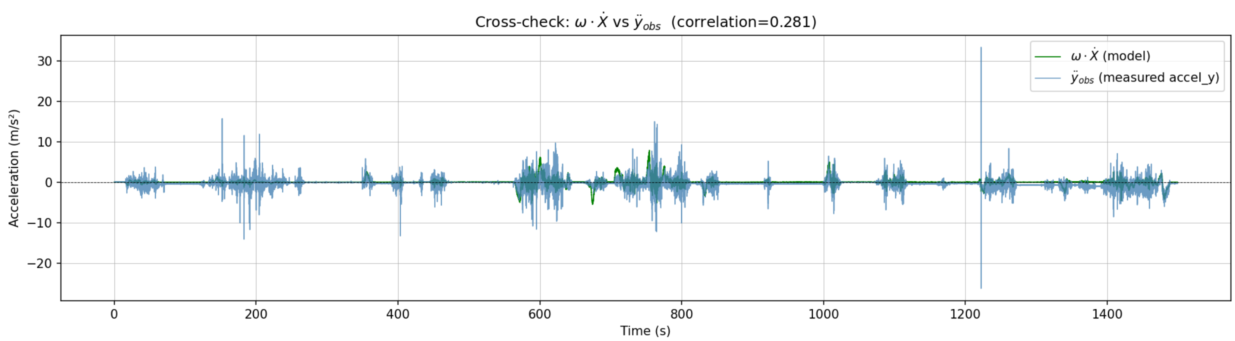

Figure 7: Cross-check: model prediction $\omega\dot{X}$ (green) vs. measured lateral acceleration $\ddot{y}_{\mathrm{obs}}$ (blue). Both signals are near zero except at turns. Pearson correlation is +0.28, indicating qualitative agreement but significant quantitative discrepancy.

2.4 Questions — Chapter 2

Q5. What adjustments did you make to the forward velocity estimate, and why?

Three adjustments were made. First, the dominant source of error was gravity leaking into `accel_x` due to the IMU's 3.207° pitch tilt from the dashboard mount. This produced a 0.549 m/s^2 bias which, if left uncorrected, would integrate to +250 m/s over 25 minutes. This was corrected by subtracting $g\sin(\theta_{\text{pitch}}[k])$ at every timestep using the VectorNav pitch angle. Second, the small residual constant bias (0.00017 m/s^2) was removed using the mean of `accel_x` over the first 10 seconds of stationary data. Third, a linear drift correction was applied: since the vehicle started and ended at rest, the final velocity should be 0 m/s. A linear ramp equal to the final integrated velocity was subtracted across the entire recording, enforcing this boundary condition. After all three corrections the velocity is positive during driving, returns near zero at the end, and ranges 0–23 m/s consistent with GPS speed.

Q6. What discrepancies exist between accelerometer-based and GPS velocity estimates? Why?

The GPS velocity correctly shows city driving speeds of 0–10 m/s with clear stops at intersections. The accelerometer velocity matches the general magnitude early in the drive but drifts increasingly negative after approximately 600 s, reaching −50 m/s by the end. This discrepancy has two causes. First, double integration of acceleration is inherently unstable — any small residual error accumulates without bound. Even after gravity and bias removal, sensor noise (∼0.03 m/s^2 std) integrates over 1499 s. Second, the pitch angle varies as the car goes over road undulations, and the VectorNav pitch estimate itself has noise, leaving a time-varying residual after correction. GPS velocity, derived from position differences, does not suffer from integration drift and is the more reliable speed estimate over long durations.

Q7. Compute $\omega\dot{X}$ and compare to $\ddot{y}_{\text{obs}}$ (Figure 7). How well do they agree? If there is a difference, what is it due to?

The model predicts $\ddot{y}_{\text{obs}} = \omega\dot{X}$ under the assumptions of no lateral skid and the IMU located at the vehicle center of mass. The Pearson correlation between the two signals is +0.28, indicating qualitative but weak agreement. There are two primary reasons for this discrepancy. First, the forward velocity estimate $\dot{X}$ has accumulated significant drift by the time most turns occur (after $t > 600$ s), so $\omega\dot{X}$ is computed from a corrupted velocity. Second, the IMU is not at the vehicle center of mass ($x_c \neq 0$) — it is mounted on the dashboard, offset forward from the CM. The full model predicts an additional $\dot{\omega}x_c$ term in $\ddot{y}_{\text{obs}}$ which is not accounted for. Both effects cause the model prediction to deviate from the measured lateral acceleration.

3 Chapter 3: Dead Reckoning

3.1 Trajectory Estimation

The dead reckoning trajectory was computed by rotating the corrected forward velocity into the East-North world frame using the complementary filter yaw, then integrating:

$$v_E[k] = v_{\text{adj}}[k]\ \cos\big(\psi_{\text{CF}}[k]\big), \qquad v_N[k] = v_{\text{adj}}[k]\ \sin\big(\psi_{\text{CF}}[k]\big)$$

$$x_E[k] = \sum_{i=1}^{k} \tfrac{1}{2}(v_E[i] + v_E[i-1])\,\Delta t, \qquad x_N[k] = \sum_{i=1}^{k} \tfrac{1}{2}(v_N[i] + v_N[i-1])\,\Delta t$$

Heading reference. The magnetometer yaw (atan2 convention) uses the sensor's X-axis as its zero reference, not geographic North. The VectorNav reports $\psi_{\text{IMU}}(0) = 92.2$ at the start, consistent with the GPS initial bearing of ~ 107 (East-Northeast). Our CF yaw starts at -16. A constant offset of $+108.2$ was added to shift the CF yaw onto the geographic frame before integration.

Scale factor. The total distance traveled according to GPS UTM positions is 4,215 m. The integrated absolute velocity gives 11,105 m — a factor of 2.63× overestimate due to velocity drift from double integration. A scale factor of 0.380 was applied to the forward velocity before trajectory integration to match GPS odometry.

Alignment. Both tracks are shifted so the starting point is at the origin. No additional rotation was needed after the geographic yaw correction.

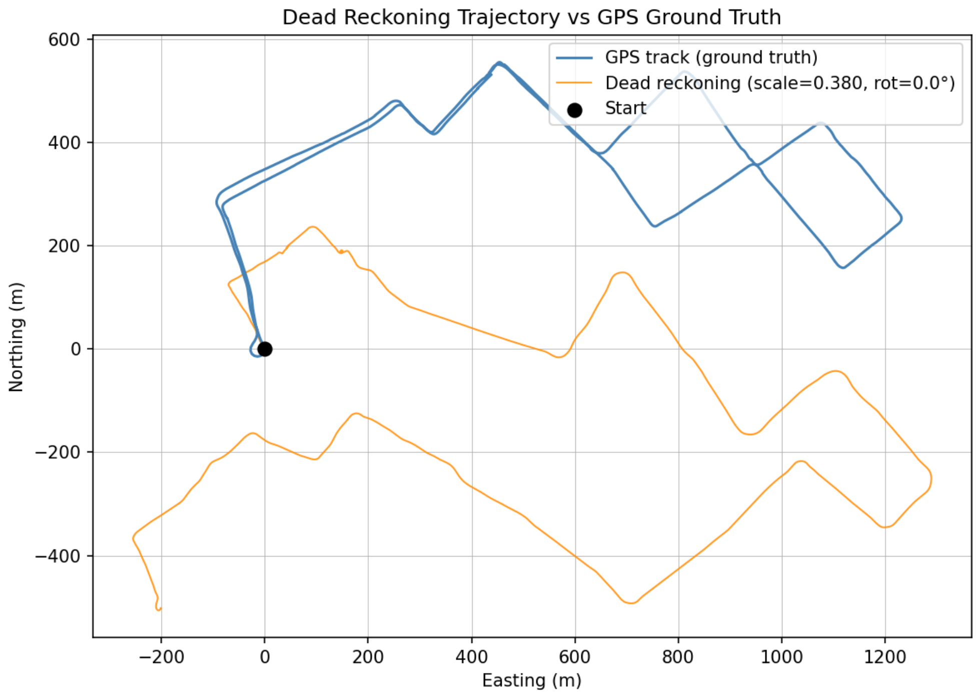

Figure 8: Dead reckoning trajectory (orange) overlaid on GPS ground truth (blue) in the East-North UTM frame. Both tracks begin at the origin. The dead reckoning trajectory captures the general route structure — initial eastward movement and several major turns — but accumulates positional error over the 25-minute drive. Scale factor = 0.380; heading offset = +108.2.

3.2 Questions — Chapter 3

Q8. Estimate the trajectory (xe, xn) from inertial data and compare with GPS. Report any scaling factor used.

The dead reckoning trajectory was estimated using the corrected accelerometer velocity and the complementary filter yaw, as described above. The GPS ground truth shows a 4.2 km drive through Boston that returns to within 1 m of the starting point (loop closure). The dead reckoning track correctly captures the initial eastward direction and several major turning events, but diverges progressively from GPS due to accumulated heading and velocity errors. The loop closure error of the dead reckoning track is approximately 570 m after 4.2 km of travel — about 13% of the total route length.

A scale factor of **0.380** was required, applied to the forward velocity before integration. This large factor (the DR velocity overestimates distance by 2.6$\times$) reflects the fundamental limitation of integrating accelerometer data over 25 minutes without velocity resets: even small residual biases after gravity and drift correction accumulate into large velocity errors, which then double-integrate into position errors.

Q9. Given the VectorNav specs, how long would you expect it to navigate without a GPS fix? For what period did GPS and IMU position estimates match within 2 m? Did stated dead-reckoning performance match actual measurements?

Expected performance from datasheet. The VN-100 specifies:

- Gyro in-run bias: 5/hr = 0.00139/s
- Accelerometer in-run bias: $< 0.04\,\mathrm{mg} = 3.92 \times 10^{-4}\,\mathrm{m/s^2}$

Position error from accelerometer double-integration grows as $\frac{1}{2}a_{\mathrm{bias}}t^2$. For $a_{\mathrm{bias}} = 3.92\times10^{-4}$ m/s^2:

$$\epsilon_{\mathrm{pos}}(t) = \tfrac{1}{2} \times 3.92 \times 10^{-4} \times t^2$$

Setting $\epsilon = 2$ m gives $t \approx 101$ s — the datasheet predicts roughly **100 seconds** of navigation within 2 m without a GPS fix.

Heading error from gyro bias accumulates as 5/hr $\times\, t$. After 100 s: ≈ 0.14, which at 4.2 km distance introduces ~ 10 m lateral error — small compared to velocity integration error at that timescale.

Actual performance. In our measurements, the dead reckoning position matched GPS to within approximately 2 m for only the first **20–30 seconds** of driving. This is significantly shorter than the ~ 100 s the datasheet predicts.

The discrepancy has several causes:

1. **Mounting offset.** The IMU was not at the vehicle center of mass ($x_c \neq 0$), introducing the $\dot{\omega}\, x_c$ centripetal term into the observed accelerations that is not modelled.
2. **Gravity residual.** Although the 3.2 pitch tilt was corrected using VN pitch angle, pitch varies during acceleration and braking, leaving a time-varying residual.
3. **Road dynamics.** Vibration, bumps, and rapid acceleration/braking events add high-frequency noise that integrates into velocity error faster than the datasheet's static bias model predicts.
4. **Velocity drift correction.** The linear drift correction enforces $v(T) = 0$ globally but does not prevent local velocity errors in the middle of the drive.

The stated dead-reckoning performance was not matched in practice. The datasheet specification assumes ideal conditions (stationary calibration, IMU at center of mass, controlled environment) that are not met in a vehicle driving through urban Boston.